

```
In [ ]: #PART 1:
#EXERCISE 1.1

my_name="Didier"
my_age=25
my_gpa=3.8
am_i_present=True

# Print the values of the variables

print("My name is:", my_name , type(my_name))
print("My age is:", my_age , type(my_age))
print("My GPA is:", my_gpa , type(my_gpa))
print("Am I present?", am_i_present , type(am_i_present))
```

```
In [ ]: #EXERCISE 1.2

age_text= "21"
age = 1+ int(age_text)

print("new result : " , age )
```

```
In [ ]: #QUESTION 1.3

print("3" + "4")

#answer will be 34 not 7 because (+) operator is used to concatenate strings
```

```
In [ ]: #PART 2:

#EXERCISE 2.1

print(f"I am {my_name} , {my_age} years old , studying at AUCA " )
```

```
In [ ]: #EXERCISE 2.2

course = "Big Data Analytics "

print (course.upper())
print(len(course))
print(course[0:4])
```

```
In [ ]: #PART 3:

#EXERCISE 3.1

scores = [72 , 85 , 91 , 78]

print(scores[0:3])
print(len(scores))
print(sum(scores) / len(scores))
print(f"maximum is {max(scores)} and minimum is {min(scores)}")
```

```
In [ ]: #EXERCISE 3.2

scores.append(88)
scores[3] = 66
print(scores)
```

```
In [ ]: #PART 4:

#EXERCISE 4.1

me = {

    "name": "Didier" ,
    "age": 25 ,
    "program " : "Day",
    "district" : "Gasaba"

}

print(me["program "])
```

```
In [ ]: #EXERCISE 4.2

me["score"]=90
me["age"]=26

print(me.keys())
print(me)
```

```
In [ ]: #PART 5:
```

```
#EXERCISE 5.1
```

```
# 1
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```
import pandas as pd
```

```
df=pd.read_excel("week1_students.xlsx")
```

```
print( "the total number of students is: ", len(df))
```

```
In [ ]: #2
```

```
f=round(df["score"].mean(),1)
```

```
print( "the average score is: ", f)
```

```
In [ ]: #3
```

```
maximum = df["score"].max()
```

```
print("the maximum score is: ", maximum)
```

```
minimum = df["score"].min()
```

```
print("the minimum score is: ", minimum)
```

```
In [ ]: #4
```

```
range = maximum - minimum
```

```
print("the range of scores is: ", range)
```

```
In [ ]: #EXERCISE 5.2
```

```
#1
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```
print(df.iloc[0])
```

```
print(df.iloc[19])
```

```
In [ ]: #2
```

```
names=df.iloc[5]["name"]
```

```
districts=df.iloc[5]["district"]
```

```
programs=df.iloc[5]["program"]
```

```
scores=df.iloc[5]["score"]
```

```
print(f"{names} from {districts} studies {programs} and scored {scores}")
```

In []: *#EXERCISE 5.3*

#1

```
gsb=df[df["district"]=="Gasabo"]
kckr=df[df["district"]=="Kicukiro"]
rwg=df[df["district"]=="Rwamagana"]

print("students from Gasabo is: ", len(gsb))
print("students from Kicukiro is: ", len(kckr))
print("students from Rwamagana is: ", len(rwg))
```

In []: *#EXERCISE 5.4*

#1

```
first_winner=df[df["score"]==df["score"].max()]

print(f"student is {first_winner.iloc[0]['name']},score:{first_winner.iloc[0]['score']}")
```

In []: *#BONUS*

```
lowest_winner=df[df["score"]==df["score"].min()]
print(f"student is{lowest_winner.iloc[0]['name']},score:{lowest_winner.iloc[0]['score']}")
```

```
In [ ]: first_winner_attendance=df[df["score"]==df["score"].max()]["attendance_pct"].values[0]
print(f"the top student attendance is {first_winner_attendance}")

lowest_winner_attendance=df[df["score"]==df["score"].min()]["attendance_pct"].values[0]
print(f"the lowest student attendance is {lowest_winner_attendance}")
```

In []: *#PART 6:*

#1

```
# in 5V's it doesn't have " VOLUME " because the data is not big enough to be considered
# as big data. The data is small and can be easily processed using traditional methods.
```

In []: *#2*

```
# I was suprised to see that with big data we can easily analze and gather
```

```
# information from dataset easily using programming languages  
# Like python and libraries like pandas. It is very easy to manipulate  
# data and get the required information from the dataset.
```